

DLD:-

Date _____ 20 _____

HOME WORK:-

Q1:-

$$\begin{aligned} \text{a) } (71.67)_8 &= 7 \times 8 + 1 \times 8^0 + \frac{6}{8} + \frac{7}{8^2} \\ &= (57.859)_{10} \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{b) } (122)_3 &= 1 \times 3^2 + 2 \times 3 + 2 \\ &= (17)_{10} \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{c) } (65A)_{11} &= 6 \times 11^2 + 5 \times 11 + 10 \times 11^0 \\ &= (791)_{10} \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{d) } (0.435)_6 &= 0 \times 6 + \frac{4}{6} + \frac{3}{6^2} + \frac{5}{6^3} \\ &= (0.773)_{10} \end{aligned}$$

10 A
11 B
12 C
13 D
14 E
15 F

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Q2:-

a) $(50.08)_4$ to base 8

$$5 \times 4 + 0 \times 1 + \frac{0}{4} + \frac{8}{4^2} = (20.5)_{10}$$

$$= (24.4)_8 \text{ Ans}$$

8	20	8
8	2	4
	0	2

b) $(0.6775)_8$ to hexadecimal $16:-$

$$= \frac{6}{8} + \frac{7}{8^2} + \frac{7}{8^3} + \frac{5}{8^4}$$

$$= (0.8742)_{10}$$

$$= (0.DFC)_{16} \text{ Ans}$$

$$\begin{array}{r} 0.5 \\ \times 8 \\ \hline 4 \end{array}$$

$$\begin{aligned} 0.8742 \times 16 &= 13.988 \\ 0.988 \times 16 &= 15.808 \\ 0.808 \times 16 &= 12.928 \end{aligned}$$

c) $(715)_{10}$ to base 13:-

13	715	
13	55	→ 0
13	4	→ 3
	0	→ 4

$$= (430)_{13}$$

d) $(110011.011)_2$ to decimal.

$$32 + 16 + 2 + 1 + \frac{1}{2^2} + \frac{1}{2^3} = (51.37)_{10}$$

Q31- 12-bit register stores $(10001001011)_2$.

a) Unique numbers:-

$$\begin{array}{cccccccccccc} & 2048 & 1024 & 512 & 256 & 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ \times 2^{11} & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}$$

$$2048 + 128 + 16 + 4 + 2 + 1 = (2199)_{10}$$

b) 2's complement :-

$$\begin{array}{cccccccccccc} & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ (-v) \text{ number} & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ & 1024 & 512 & 256 & 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \end{array}$$

(1's complement)
(2's complement)

convert-

$$1024 + 512 + 256 + 64 + 32 + 8 + 1 = (-1897)_{10}$$

c) Ori complement-

$$8 + 32 + 64 + 256 + 512 + 1024 = 1896$$

$$= (1896)_{10}$$

$$1's \text{ complement} = -1896$$

d)

$$\begin{array}{ccc} 1000 & 1001 & 0111 \\ 8 & 9 & 7 \end{array} = [897 \text{ BCD}]$$

	Actual		Coded
Q4:-	0	=	@
a)	1	=	*
	2	=	\$
	3	=	%
	4	=	^
	5	=	/
	6	=	<
	7	=	>
	8		*@
	9		**
	10		*\$
	11		*%
	12		*^

b) $(1601)_{10}$

$1 = *$ $6 = <$ $0 = @$ $1 = *$

so year 1601 = $* < @ *$ in the coded form

c) $\$@* = (201)$ in ^{decimal} ~~binary~~ ^{digit}.

Working for Question #5:-

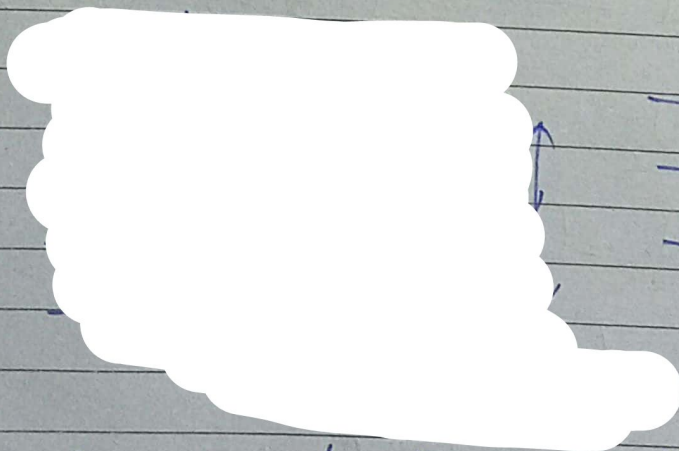
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Q5:- $(975)_{10}$ Converting directly:-

2	975	
2	487	→ 1 ↑
2	243	→ 1
2	121	→ 1
2	60	→ 1
2	30	→ 0
2	15	→ 0
2	7	→ 1
2	3	→ 1
2	1	→ 1
2	0	
	Quotient	
	Remainder	

$$(975)_{10} = (\text{111001111})_2$$

- Converting to hexa decimal:-



16	975	
16	60	→ 15 ↑
16	3	→ 12
	0	→ 3

3 12 15
3 C F

$$(3CF)_{16} = (\text{111001111})_2$$

Answer for Q5:-

Direct conversion method requires ~~extensive~~ extensive integer continuous division especially integer computational power will be required for larger numbers.

- But first converting it into hexadecimal will require less number of divisions and just looking up a table for converting from hexadecimal to binary. Hence second way is much more efficient.

Q6(a)

1	1	6	6	11	B	16	G
2	2	7	7	12	C	17	H
3	3	8	8	13	D	18	I
4	4	9	9	14	E	19	J
5	5	10	A	15	F	20	K

b) B has to repay the amount of $\rightarrow (DHJSF)_{16}$

+ Interest of $\rightarrow (21DHH)_{16}$

He repayed in 8 months, that means

he was 2 months late so he

has to pay additional $\rightarrow (5B3E) + (5B3E)$

~~Convert~~

$$(DHJSF)_{16} = F(2223715)_{10}$$

$$(21DHH)_{16} = (333557)_{10}$$

$$(5B3E)_{16} = (44474)_{10}$$

$$\text{Total sum} = 2223715 + 333557 + (44474 \times 2) = 2646220$$

Total amount = 264 6220

20	2646220	
20	132311	→ 0
20	6615	→ 11
20	330	→ 15
20	16	→ 10
	0	→ 16

$(9AFBO)_{10}$ = Total amount that should be paid back.

The net gain = $(333557)_{10} + (44474)_{10} \times (2)_{10} = (422505)_{10}$

in decimal:- $(110011001001101001)_2$

2	422505		2	206	
2	211252	→ 1	2	103	→ 0
2	105626	→ 0	2	51	→ 1
2	52813	→ 0	2	25	→ 1
2	26406	→ 1	2	12	→ 1
2	13203	→ 0	2	6	→ 0
2	6601	→ 1	2	3	→ 0
2	3300	→ 1	2	1	→ 1
2	1650	→ 0		0	→ 1
2	825	→ 0			
2	412	→ 1			
	206	→ 0			

Q 7:-

last three alphabets of Sualah = LEH

first three alphabets of last Name. Rizvi = RIZ

$$X_1 = LEH = 2C2528$$

$$X_2 = RIZ = 32293A$$

$$a) X_1 - X_2 = X_1 + 16's \text{ complement of } X_2$$

$$X_2 :- 16's \text{ complement} :- \begin{array}{r} F F F F F F \\ 3 2 2 9 3 A \\ \hline C D D 6 C 5 + 1 = (C D D 6 C 6)_{16} \end{array}$$

Now add with X_1 :-

$$\begin{array}{r} \text{No carry} \quad 2 C 2 5 2 8 \\ + \quad C D D 6 C 6 \\ \hline F 9 F B E E \end{array}$$

as there is no carry that means the Answer is negative
so, take 16's complement again:-

$$\begin{array}{r} F 9 F B E E \quad 16's \text{ complement} = F F F F F F \\ F 9 F B E E \\ \hline 0 6 0 4 1 1 + 1 \\ = \\ (-060412)_{16} \end{array}$$

5) $X_2 - X_1 = X_2 + X_1's \text{ 16's complement}$

$X_1's \text{ 16's complement}$

$$\begin{array}{r} F F F F F F \\ 2 C 2 5 2 8 \\ \hline (D 3 D A D 7)_{16} = (D 3 D A D 7)_{16} \end{array}$$

Now:- $X_2 + X_1's \text{ complement}$

$$\begin{array}{r} \begin{array}{c} \nearrow \\ + \end{array} \begin{array}{r} 3 \ 2 \ 2 \ 9 \ 3 \ A \\ D \ 3 \ D \ A \ D \ 7 \\ \hline 0 \ 6 \ 0 \ 4 \ 1 \ B \end{array} \quad \begin{array}{l} 13+3=16-16=0 \\ 13+3=16-16=0 \\ 17-16=1 \end{array} \end{array}$$

carry over if it is possible.

$(060411)_{16}$ is the answer.

c) To convert to octal

for $X_1 = 2C2528$

$$2 = 0010$$

$$C = 1100$$

$$2 = 0010$$

$$5 = 0101$$

$$2 = 0010$$

$$8 = 1000$$

$$2C2528 = \underline{0010} \underline{1100} \underline{0010} \underline{0101} \underline{0010} \underline{1000}$$

Now convert each term group into octal as $2^3 = 8$

$$001 = 1$$

$$011 = 3$$

$$000 = 0$$

$$010 = 2$$

$$010 = 2$$

$$100 = 4$$

$$100 = 4$$

$$000 = 0$$

$$X_1 = (13022450)_8$$

for X_2 :- 3 2 2 9 3 A

0011001000100100111010

$$001 = 1$$

$$100 = 4$$

$$100 = 4$$

$$010 = 2$$

$$100 = 4$$

$$100 = 4$$

$$111 = 7$$

$$010 = 2$$

$$X_2 = (14424472)_8$$

Sum:-

$$\begin{array}{r} 13022450 \\ + 14424472 \\ \hline 27447192 \end{array}$$

Sum:- $X_1 \& X_2$ in Octal radix = $(27447192)_8$



Q8:-

$$a) \quad a = 11043$$

$$b = 3289$$

$$F-3 = C$$

$$F-2 = D$$

$$F-8 = 7$$

$$F-4 = 6$$

$$r's \text{ complement} = a + b's \text{ complement}$$

$$b's \text{ complement} = CD76 + 1 = CD77$$

$$b) (11001.101)_2 - (10111.001)_2$$

$$a = 11001.101$$

$$b = 10111.001$$

$$a - b = a + b's \text{ 2's complement}$$

b's complement-

$$\begin{array}{r} 11111.111 \\ - 10111.001 \\ \hline 01000.110 \end{array}$$

Now

a + b's

$$\begin{array}{r} 11001.101 \\ + 01000.110 \\ \hline 00010.011 \end{array}$$

Carry that goes to the number is positive

$$11001.101 - 10111.001 = 10.011$$

$$11043$$

$$+ CD77$$

$$\hline 2DD19$$

Now

$$FFFFF$$

$$- 1DD19$$

$$\hline E22E6$$

Ans

Q9:- Convert the following to BCD.

a) $0.493 + 1.965$

$$0.493 = 0.0100\ 1001\ 0011$$

$$1.965 = 0001.1001\ 0110\ 0101$$

$$\begin{array}{r}
 1.1001\ 0110\ 0101 \\
 + 0.0100\ 1001\ 0011 \\
 \hline
 10.1110\ 1111\ 1000 \\
 + 0110\ 0110 \\
 \hline
 10.0100\ 0101
 \end{array}$$

$$\begin{array}{r}
 \cancel{1110\ 0101\ 1000} \\
 10.0100\ 0101\ 1000 \quad \checkmark
 \end{array}$$

b) $79.84 - 34.51$

$$79.84 = 0111\ 1001\ 1000\ 0100$$

$$34.51 = 0011\ 0100\ 0101\ 0001$$

$$\begin{array}{r}
 0111\overset{\circ}{X}001, X\overset{\circ}{X}\overset{\circ}{X}\overset{\circ}{0} 0X\overset{\circ}{X}\overset{\circ}{0} \\
 - 0011 0100, 0101 0001 \\
 \hline
 0100 0101, 0011 0011
 \end{array}$$

$$1000 101, 0011 0011$$

c) $1357 + 870$

$$\begin{array}{r}
 1357 = 0001' 0'011' 0'1'01 0111 \\
 870 = + 1000 0111 0000 \\
 \hline
 0000 1100 \quad | \quad 1100 \quad | \quad 0111 \\
 + 0110 \quad | \quad 0110 \\
 \hline
 0010 \quad | \quad 10010
 \end{array}$$

~~$$0001100 0010 0111$$~~

$$1357 + 870 = (\text{BCD}) 0010 0010 0010 0111$$

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d) $0.721 + 1.48$

$$\begin{array}{r}
 \begin{array}{cccc}
 \overset{\text{1}}{\underset{\text{1}}{\text{0000}}} & \overset{\text{1}}{\underset{\text{1}}{\text{0011}}} & \text{0010} & \text{0001} \\
 + & \text{0001} & \text{0100} & \text{1000} \\
 \hline
 \text{0010} & \text{0110} & \text{1101} & \text{0001}
 \end{array} & \begin{array}{l} (0.721) \\ (1.480) \end{array} \\
 \begin{array}{cc}
 & \text{1100} \\
 & \text{0110} \\
 \text{10010} & \text{0000}
 \end{array}
 \end{array}$$

$0010.0010 \text{ } 000000001$

$$a) (+12)_{10} + (-1)_{10}$$

question 10

$$\downarrow$$

$$(001100)_2$$

$$\downarrow$$

$$(000001)_2$$

Take 2's complement

$$111110 + 1 = 111111$$

Now perform addition-

$$\begin{array}{r} \cancel{001100} \quad \cancel{000001} \\ + \\ \hline 1001011 \end{array}$$

It is not a overflow

~~It is a overflow.~~

$$b) (-7)_{10} - (-8)_{10} = -7 + 8 \rightarrow$$

$$\downarrow$$

$$000111$$

2's complement

$$111000$$

$$001000$$

Now

$$111000 + 001000$$

$$\boxed{000001}$$

not a overflow.

$$c) (10)_{10} - (+30)_{10} = 10 + (-30)_{10}$$

$$\downarrow$$

$$000010$$

$$\downarrow$$

$$011110$$

$$(-ve) 100001 = 100001$$

add

$$\begin{array}{r} 000010 \\ \oplus 011110 \\ \hline 000000 \end{array}$$

$$\rightarrow 101100$$

No overflow.



Q12

Decimal	Signed Magnitude	1's complement	2's complement
-8108	11101100	10010011	10010100
-128	1 0000000	N/A	10000000
-23	10010111	11101000	11101001
50	00110010	00110010	00110010
122	01111010	01111010	01111010
0	00000000	00000000	00000000

Q13:

i- 00010110

• as we have ^{even} ~~odd~~ number of 1's i.e:- 3 so we need 1 as our even parity bit.

• So number becomes: 100010110

ii- 10100101

• as we need to have even number of 1's so we already have 4 1's, so our even parity digit is zero.

• So the number becomes 010100101.

Q14.

a - to excess 3-

1001	0101	0011	0111
↓	↓	↓	↓
9	5	3	7
+ 3	3	3	3
12	8	6	10
↓	↓	↓	↓
1100	1000	0110	1010

So, Number converts to 1100 1000 0110 1010

b) Binary to gray code:-

0110110001011000

1st bit ~~0~~ = 02nd bit = $1 \text{ XOR } 0 = 1$ 3rd bit = $1 \text{ XOR } 1 = 0$ 4th bit = $1 \text{ XOR } 1 = 0$ 5th bit = $0 \text{ XOR } 1 = 1$ 6th bit = $1 \text{ XOR } 0 = 1$ 7th bit = $1 \text{ XOR } 1 = 0$ 8th bit = $0 \text{ XOR } 1 = 1$ 9th bit = $0 \text{ XOR } 0 = 0$

10th bit = " = 0

11th bit = $1 \text{ XOR } 0 = 1$ 12th bit = $0 \text{ XOR } 1 = 1$ 13th bit = $1 \text{ XOR } 0 = 1$ 14th bit = $1 \text{ XOR } 1 = 0$ 15th bit = $0 \text{ XOR } 0 = 0$ 16th bit = $0 \text{ XOR } 0 = 0$

So number Gray code =

0100110100110000